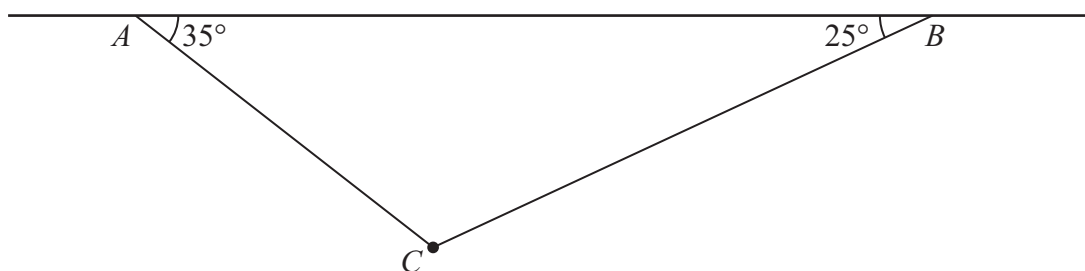


(3)

(3)

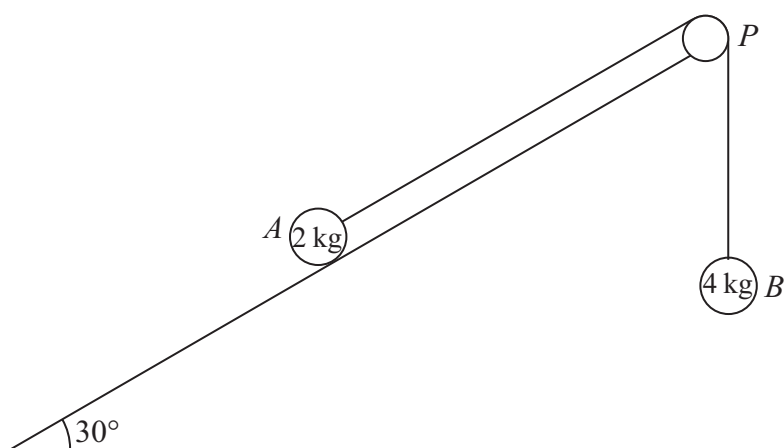


A particle of weight 8 N is attached at C to the ends of two light inextensible strings AC and BC . The other ends, A and B , are attached to a fixed horizontal ceiling. The particle hangs at rest in equilibrium, with the strings in a vertical plane. The string AC is inclined at 35° to the horizontal and the string BC is inclined at 25° to the horizontal, as shown in Figure 1. Find

- (i) the tension in the string AC ,
- (ii) the tension in the string BC .

(8)





A fixed rough plane is inclined at 30° to the horizontal. A small smooth pulley P is fixed at the top of the plane. Two particles A and B , of mass 2 kg and 4 kg respectively, are attached to the ends of a light inextensible string which passes over the pulley P . The part of the string from A to P is parallel to a line of greatest slope of the plane and B hangs freely below P , as shown in Figure 2. The coefficient of friction between A and the plane is $\frac{1}{\sqrt{3}}$. Initially A is held at rest on the plane. The particles are released from rest with the string taut and A moves up the plane.

Find the tension in the string immediately after the particles are released.

(9)



(5)

(2)

5.

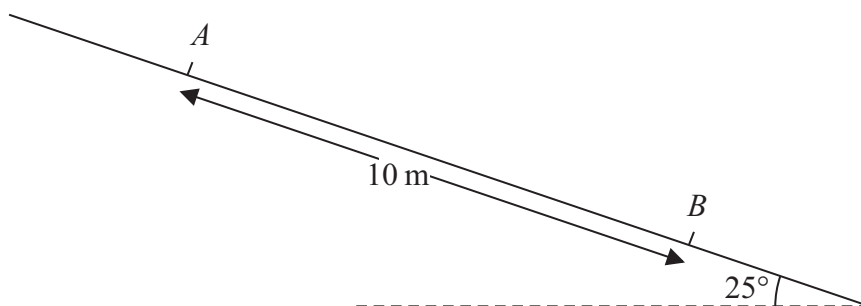


Figure 3

A particle P of mass 0.6 kg slides with constant acceleration down a line of greatest slope of a rough plane, which is inclined at 25° to the horizontal. The particle passes through two points A and B , where $AB = 10 \text{ m}$, as shown in Figure 3. The speed of P at A is 2 m s^{-1} . The particle P takes 3.5 s to move from A to B . Find

- (a) the speed of P at B , (3)
- (b) the acceleration of P , (2)
- (c) the coefficient of friction between P and the plane. (5)



(a) Find the position vector of S at time t hours. (2)

(b) Find the value of n . (5)

(c) Find the distance OP . (4)

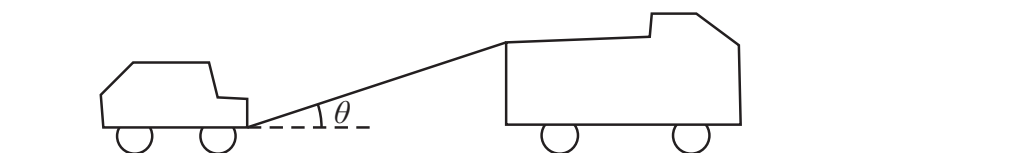


Figure 4

(a) Find the deceleration of the truck and the car.

(3)

(b) the force in the towbar,

(4)

(c) the value of R .

(4)



8.

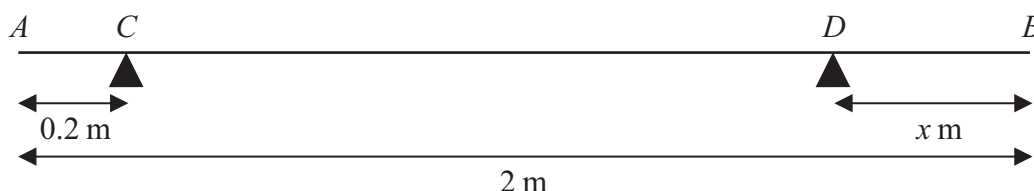


Figure 5

A uniform rod AB has length 2 m and mass 50 kg. The rod is in equilibrium in a horizontal position, resting on two smooth supports at C and D , where $AC = 0.2$ metres and $DB = x$ metres, as shown in Figure 5. Given that the magnitude of the reaction on the rod at D is twice the magnitude of the reaction on the rod at C ,

- (a) find the value of x .

(6)

The support at D is now moved to the point E on the rod, where $EB = 0.4$ metres. A particle of mass m kg is placed on the rod at B , and the rod remains in equilibrium in a horizontal position. Given that the magnitude of the reaction on the rod at E is four times the magnitude of the reaction on the rod at C ,

- (b) find the value of m .

(7)

